

PART II INDIAN GEOGRAPHY

India occupies a South-Central position in the Asian continent, looking across the Arabian sea to Arabia and Africa on the West and across Bay of Bengal to Myanmar, Malaysia and the Indonesian Archipelago on the East. Geographically, the Himalayan ranges keep India apart from the rest of Asia. India derives her name from river Indus.

Location

India is often described as a tropical country although a large part of India lies in sub-tropics. The territorial limits of the Indian mainland extend between 8° 4' N and 37° 6' N latitudes and 68° 7' E and 97° 25' E longitudes.

Area and Extent

- India ranks 7th in the world in terms of area after Russia, Canada, USA, China, Brazil and Australia.
- It has a total land area of about 3287263 sq km, i.e. about 33 lakh sq km.
- It is second largest in terms of population and holds 17.4% of the total world population, the land area of India amounts only to 2.4% of the total world landmass.
- India is the second largest country in Asia both in terms of area as well as population, after China.

Country	States
Pakistan	Jammu and Kashmir, Punjab, Rajasthan, Gujarat
China	Jammu and Kashmir, Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh
Nepal	Uttarakhand, Uttar Pradesh, Bihar, West Bengal, Sikkim
Bhutan	Sikkim, West Bengal, Assam, Arunachal Pradesh
Myanmar	Arunachal Pradesh, Nagaland, Manipur, Mizoram
Bangladesh	West Bengal, Meghalaya, Assam, Tripura, Mizoram

- State with longest coastline is **Gujarat**.
- Active volcanoes are in **Barren island in Andaman and Nicobar islands**.
- Northern most point of India is Indira Col.
- Southern most point is **Indira point or Pygmalion point in Great Nicobar**.
- Southern most tip of mainland is **Kanyakumari**.
- Eastern most point is Kibithu, Arunachal Pradesh.
- Western most point is Guhar moti in Kutch, Gujarat.

Indian Standard Time (IST)

- India has only one standard time.
- India is 5.5 hours ahead of GMT/UTC, 4.5 hours behind Australian Eastern standard time and 10.5 hours ahead of American Eastern standard time.
- The $82\frac{1}{2}^{\circ}$ E longitude that passes through Allahabad city is chosen as standard longitude for Indian standard time.

RELIEF AND PHYSIOGRAPHIC DIVISIONS

Physiography is the branch of geography, which studies the present relief features of the Earth's surface or of natural features in their causal relationships. The physiographic diversity of India embraces fold mountains, flat plains and one of the oldest plateaus of the world.

India is divided into five physiographic units :

1. The Himalayas

The Northern mountain wall is a series of high mountain ranges stretching over the Northern borders of India.

- The geologically young and structurally fold mountain ranges, the Himalayas run in West-East direction from the Indus to the Brahmaputra.
- They form an arc, which covers a distance of about 2400 km. Their width varies from 400 km in Kashmir to 150 km in Arunachal Pradesh.
- The altitudinal variations are greater in the Eastern half than those in the Western half.

The Himalayas range is classified into four longitudinal series of mountains, which are as follow

- Trans-Himalayas** North of the Greater Himalayas lie the Trans-Himalayas or the Tibet Himalayas. This section is older than Himalayas. This range acts as a watershed between rivers flowing towards South and those flowing towards North. These ranges are about 40 km wide and rise in height upto 5000 m. They include the Karakoram, Zaskar and Ladakh ranges.
- Greater Himalayas or Himadri** The Northern most important range is known as the Great or Inner Himalayas or the Himadri.
- Himachal Himalayas** The range lying to the South of the Himadri, forms the most rugged mountain system and is known as Himachal or Lesser Himalayas or Middle Himalayas.

- iv. **Shivaliks** The outermost range of the Himalayas is called the **Shivaliks**. They extend over a width of 10-50 km and have an altitude varying between 900 and 1100 m. These ranges are composed of unconsolidated sediments brought down by rivers from the main Himalayan ranges located in North.
- v. **The Purvanchal** After crossing the Dihang Gorge, the Himalayas take a sudden Southward turn and form a series of comparatively low hills in the shape of crescent with its convex side pointing towards the West. These hills are known as Purvanchal.

IMPORTANT PEAKS

- Highest mountain peak in India K2 or Godwin Austin (PoK).
- Highest peak of India in Himalaya is Kangchenjunga.
- Highest peak in Eastern Ghats Arma Konda (AP).
- Highest mountain peak in Western ghats Annaimudi.
- Highest peak in Aravali is Gurushikhar in Mount Abu.
- Highest peak in Satpura and Mahadeo hills Dhupgarh.
- Highest peak in Nilgiris Doda Betta.
- Highest peak in Andaman and Nicobar Islands Saddle Peak.
- Highest peak of Naga hills Saramati Peak.

Name	State	Features
Rohtang pass	Himachal Pradesh	It is a high mountain pass on Eastern Pir Panjal range of the Himalayas around 51 km from Manali. It connects the Kullu of Himachal Pradesh, India.
Shipki La	Himachal Pradesh	The river Sutlej enters India through this pass.
Jelep La	Sikkim	Jelep La is a high mountain pass between India and Tibet in East Sikkim District of Sikkim. The famous Menmecho lake lies below the Jelep La pass.
Nathu La	Sikkim	It connects the Indian State of Sikkim with China's Tibet autonomous region.
Lipulekh pass	Uttarakhand	It is a Himalayan pass connecting the Kumaon region of Uttarakhand in the Pithoragarh district in India with the old trading town of Talakot in Tibet.

2. The Great Indian Plain

- To the South of Himalayas and North of Peninsula lies the Great Plain of North India. It is an aggradational plain formed mainly by the work of 3 rivers systems viz, the Indus, the Ganga and the Brahmaputra.
 - This is the largest alluvial tract of the world, extending for a length of 3200 km and width varies between 150 to 300 km.
- Regional division of Great Plain of India are as follows :*

Punjab-Haryana Plain

- The western part of the Northern Plain is known as Punjab-Haryana Plain. Its Eastern boundary in Haryana is formed by the Yamuna river. It also includes North-Eastern part of Rajasthan.
- Depositional processes by the rivers continuing since long, has united these doabs. However, this mass of alluvium is broken by bluffs, locally known as Dhayas.

Ganga Plain

This is the largest unit of Great Plain of India. Depending upon its geological variations, *this plain can be further sub-divided into the following three divisions :*

(i) Upper Ganga Plain

- Compacting the upper part of Ganga plain, this plain is delimited by 300 m contour in Shiwaliks in the North, the Peninsular body in the South and course of Yamuna river in the West and 100 m contour in East.
- The gradient is comparatively steeper in the North.

(ii) Middle Ganga Plain

- To the East of upper Ganga plain, lies middle Ganga plain occupying Eastern part of Uttar Pradesh and Bihar.
- This plain is drained by the Ghaghara, the Gandak and the Kosi rivers.

Name	State	Features
Banihal pass	Jammu and Kashmir	Banihal pass is a pass across the Pir Panjal range at 2832 m. This mountain range separates the Kashmir valley in the Indian state of Jammu and Kashmir from the outer Himalayas and plains to the South.
Changla pass	Jammu and Kashmir	Highest mountain pass in Ladakh. The Changla is on the route to Pangong lake from Leh.
Khardung La	Jammu and Kashmir	Khardung La is historically important as it lies on the major caravan route from Leh to Kashgar in Central Asia.
Namika La	Jammu and Kashmir	Namika La is one of two high passes between Kargil and Leh, the other is the even higher Fotu La pass.
Zoji La pass	Jammu and Kashmir	Zoji La is a high mountain pass in India, located on the Indian National Highway-1 between Srinagar and Leh in the Western sections of mountain range.
Bara-lacha La	Himachal Pradesh	Also known as Bara-lacha pass. Highest mountain pass in Zaskar range connecting Lahaul district in Himachal Pradesh to Ladakh in Jammu and Kashmir.

- Major unit of this plain are Ganga-Ghaghara doab, Ghaghara-Gandak doab and Gandak-Kosi doab (Mithila plain).

(iii) Lower Ganga Plain

- Some districts of Bihar and whole of West Bengal are part of this plain. The Northern part of this plain has been formed by sediment deposited by the Tista, Jaldhaka and Torsa.
- This area is marked by drawn and barren plain, a tract of old alluvium between Kosi-Mahananda corridor in the West and the river Sankosh in the East.
- The delta formation accounts for about two-thirds of this plain. This is the largest delta in the world. Large part of the coastal deltas is covered by thick inaccessible tidal forests called **Sunderbans**.

Brahmaputra Plain

- Western boundary of these plains are formed by Indo-Bangladesh border as well as boundary of lower Ganga plain. The Brahmaputra river enters this plain near Sadiya and flows further to Bangladesh after turning Southwards near Dhubri.
- There are large marshy tracts in this region and Southern tributaries of Brahmaputra also have meandering course and there are good number of bails and ox-bow lakes.

3. The Peninsular Plateau

- The Peninsular plateau is a tableland composed of mainly Archean gneisses and schists. It was formed due to the breaking and drifting of the Gondwana land and thus making it a part of the oldest landmass. This region of the country is surrounded on three sides by water and thus is a Peninsular plateau.
- The plateau has broad and shallow valleys and rounded hills. Narmada river, which flows into a Rift

valley, divides the region into two parts namely, the Central Highlands in its North and the Deccan plateau in its South.

Plateaus of Peninsular India

The Deccan Plateau

- It is the largest unit of the Peninsular plateau of India with an elevation of 600 m. It is higher in the South than in its North.
- It generally slopes from West to East and various big rivers of Southern India like Godavari, Krishna, Kaveri etc flow through it.
- Karnataka plateau with Archean formations lies to the South of Maharashtra plateau having rocks of lava origin in Northern Karnataka called **Malnad region** and the rest of the red soil region of the plateau called **Maidan**.

Meghalaya Plateau

- It is extension of peninsular plateau, which has been separated by huge fault. Fault is between Rajmahal hills and Meghalaya plateau.

The Chotanagpur Plateau

- It lies East of Baghelkhand in the state of Jharkhand covering some parts of Chhattisgarh and West Bengal. Its average elevation is 700 m above sea level.
- It is the storehouse of minerals and a large scale mining of iron, manganese, coal, uranium etc is done in this region.
- Damodar river valley is well-known for its coal deposits.

The Malwa Plateau

- Largely in Western Madhya Pradesh and South-Eastern Rajasthan forms a triangular shape and is typical for having two systems of drainage.

The Marwar Uplands

- The Marwar uplands of Eastern Rajasthan lie to the East of Aravalli ranges. They are made up of sandstones and limestones of the Vindhyan period.

Hill Ranges of Peninsula

The Aravalli Ranges

- It runs North-East to South-West for 800 km from Delhi through Rajasthan to Palanpur in Gujarat.
- It has a lower elevation between Delhi and Ajmer, where it is characterised by a chain of discontinuous ranges.
- Gurushikhar (1722 m) is the highest peak of the range, located in Abu hills of Rajasthan.

The Vindhyan Ranges

- It runs parallel to the Narmada Rift valley as an escarpment in an East-West direction from Jabat in Gujarat to Sasaram in Bihar for a distance of 1200 km.

The Satpura Ranges

- It is a series of seven mountains that run in the East-West direction in between Narmada and Tapi rivers. It is an example of block mountain.
- Amarkantak is meeting point of Vindhyan and Satpura range.

The Eastern Ghats

- These are discontinuous and irregular and dissected by rivers draining into the *Bay of Bengal*.
- The Eastern ghats stretch from the South of Mahanadi valley to the Nilgiris in the South. The Eastern ghats are comparatively broader and do not form a continuous water divide.

The Western Ghats or Sahyadris

- Sahyadris form the Western edge of the Deccan plateau and lie parallel to the Western coast. They form a continuous water divide.
- It runs continuously for 1600 km from Maharashtra to Kanyakumari and can be crossed through passes only.
- Highest mountain peak in Western ghat is Annaimudi. The Western ghats are higher than the Eastern ghats.

4. The Coastal Plains

The Peninsular plateau is flanked by stretch of narrow coastal strips, running along the Arabian sea on the West and the Bay of Bengal on the East.

Eastern Coastal Plains

- The plains along the Bay of Bengal, i.e. the **Eastern coastal plains** are wide and level.
- In the Northern part, they are referred to as the Utkal plains and Northern Circars, while the Southern part is known as the **Coromandel coast**.

Western Coastal Plains

- It run almost parallel between Sahyadris and Arabian sea, from Kanyakumari to Surat.
- North to South it has been classified as Kathiawar plain, Konkan plain, Karnataka plain and Malabar plain along the Kerala coast.

Rajasthan Desert

- Also known as **Thar** or **Great Indian desert**, which covers Western Rajasthan and the adjoining part of Pakistan. Desert is called **Marusthali**.
- The Eastern part of the Marusthali is rocky, while its Western part is covered by shifting sand dunes locally called dhrian.
- The Eastern part of Thar desert upto Aravalli range is semi-arid plain, which is known as **Rajasthan Bagar**. It is drained by a number of seasonal streams creating fertile tracts locally known as **Rohi**.

5. The Islands

Apart from the large number of islands in the near proximity of the Indian coast, there are two main groups of islands in the Indian Ocean far away from the coast.

Lakshadweep Islands

- These islands group lies close to the Malabar coast of Kerala. This group of 25 islands is composed of small coral islands.

- The islands North of 11° N latitude are known as **Amindivi islands** and those South of it are **Cannanore islands**.

Andaman and Nicobar Islands

- Andaman and Nicobar archipelago has been formed by the extension of the tertiary mountain chains of Arakan yoma. These islands lie close to equator and experience equatorial climate and have thick forest cover. Some of the islands are fringed with coral reefs.
- The entire group of islands is divided into two broad categories—The Andamans in the North and the Nicobars in the South. The Great Andaman group of islands in the North is separated by the **Ten Degree Channel** from the Nicobar group in the South.

DRAINAGE SYSTEM OF INDIA

- India is blessed with hundreds of large and small rivers, which drains the length and breadth of the country. Water drains in two direction of the main water divide line of India. 90% of land water drains into Bay of Bengal and the rest drains into Arabian sea.
- *The Indian rivers are divided into following two major groups*
 - i. The Himalayan Rivers
 - ii. The Peninsular Rivers

The Himalayan Rivers

The Himalayan river system is divided into three major river system, which are as follow

The Indus River System

- The Indus, also known as **Sindhu**, is the Western most of Himalayan rivers in India. It is one of the largest river basins of the world covering an area of 1165000 sq km (in India it is 321289 sq km) and a total length of 2880 km (in India 1114 km).

- It originates from a glacier near **Bokhar Chu** in the Tibetan region near Mansarovar lake. In Tibet, it is known as **Singi Khamban** or Lion's mouth.
- In Jammu and Kashmir, its Himalayan tributaries are Zaskar, Dras, Gartang, Shyok, Shingar, Nubra, Gilgit etc. Its most important tributaries, which join Indus at various places, are Jhelum, Chenab, Ravi, Beas and Sutlej.

The Ganga River System

- The Ganga system is the second major drainage system of India. It rises in the Gangotri glacier near Gaumukh (3900 m) in the Uttarakhand. Here, it is known as the Bhagirathi. At Devprayag, the Bhagirathi, meets the Alaknanda, hereafter, it is known as the Ganga.
- The Alaknanda has its source in the Satopanth glacier above Badrinath. The Alaknanda consists of the Dhauti and the Vishnu Ganga, which meet at Joshimath or **Vishnu Prayag**. The other tributaries of Alaknanda such as the Pindar joins it at **Karna Prayag**, while Mandakini or Kali Ganga meets it at **Rudra Prayag**.
- The left bank tributaries of Ganga are Ramganga, Gomti, Kali or Sharda, Gandak, Kosi and Mahanadi. The right bank tributaries of Ganga are Yamuna and Son. Yamuna joins the Ganga at Allahabad.
- Kosi is called as 'Sorrow of Bihar' while Damodar is called as 'Sorrow of Bengal' as these cause floods in these regions. Hooghly is a distributory of Ganga flowing through Kolkata.

The Brahmaputra River System

- It is one of the largest river of the world. It is known as Tsangpo in Tibet, Dihang or Siang in Arunachal Pradesh, Brahmaputra in Assam and Jamuna in Bangladesh.

- Brahmaputra forms largest number of riverine islands. **Majuli** is the largest riverine island in the world. The combined stream of Ganga and Brahmaputra forms the biggest delta in the world, the Sunderbans, covering an area of 58752 sq km. Its major part is in Bangladesh.
- Brahmaputra is volume wise largest river of India, whereas lengthwise Ganga is the longest in India. Tributaries of river are—Manar, Subanshri, Dibang and Lohit.

The Peninsular River System

- The Peninsular drainage system is older than the Himalayan one. A large number of the Peninsular rivers are seasonal, as their flow is dependent on rainfall.
- Most of the major rivers of the Peninsula such as Mahanadi, Godavari, Krishna and Cauvery flow Eastwards and drain into the Bay of Bengal. These rivers make deltas at their mouths.
- Narmada and Tapi are the only long rivers, which flow Westward and make estuaries rather than making a delta because of their swift flow and steep slopes. The drainage basins of the Peninsular rivers are comparatively smaller in size.

Lakes

A lake is an area of variable size that is filled with water and surrounded by land. Lakes form due to receding glaciers, plate tectonics, volcanism, meandering rivers, landslides and human damming. Man-made lakes are referred as reservoirs. *Facts related to Indian lakes are given below*

- Wular lake (Jammu and Kashmir) is the largest fresh water lake in India. Tulbul Navigation Project is located on this.
- Sambhar, Didwana, Lunkaransar, Panchpadra are some of the important saline lakes of Rajasthan.
- Ukai reservoir (Gujarat) is a man-made lake on river Tapi. Ranapratap Sagar and Jawahar Sagar (Rajasthan) and Gandhi Sagar (Madhya Pradesh) are located on river Chambal. Govind Sagar (Himachal Pradesh) is a huge reservoir.
- Nagarjuna Sagar (Andhra-Telangana) on river Krishna, Nizam Sagar (Telangana) on Manjra and Tungabhadra (Karnataka) on Tungabhadra are man-made lakes.
- Govind Ballabh Pant Sagar (Chhattisgarh-Uttar Pradesh) has been constructed on Rihand. Stanley reservoir is located in Tamil Nadu on Kaveri.
- Loktak lake (Manipur) is the largest fresh water lake in North-Eastern India. Keibul Lamjao is a floating national park located inside the lake.
- Chilika lake (Odisha) is the largest lagoon in India. Kolleru lake (Andhra Pradesh) is a lake located around delta region. Pulicat lake (Andhra Pradesh) is a lagoon.
- Vembanad is located in Kerala. Lonar lake is a crater lake located in Buldhana, Maharashtra.

Name of Dam	State	River
Srisailem Dam	Andhra Pradesh	Krishna River
Ukai Dam	Gujarat	Tapti River
Pandoh Dam	Himachal Pradesh	Beas River
Bhakra Nangal Dam	Himachal Pradesh and Punjab Border	Sutlej River
Nathpa Jhakri Dam	Himachal Pradesh	Sutlej River
Chamera Dam	Himachal Pradesh	Ravi River
Baglihar Dam	Jammu and Kashmir	Chenab River
Uri Hydroelectric Dam	Jammu and Kashmir	Jhelum River
Maithon Dam	Jharkhand	Barakar River
Panchet Dam	Jharkhand	Damodar River
Tungabhadra Dam	Karnataka	Tungabhadra River
Krishna Raja Sagara Dam	Karnataka	Kaveri River
Idukki Dam	Kerala	Periyar River
Parambikulam Dam	Kerala	Parambikulam River
Mullaperiyar Dam	Kerala	Periyar River
Bargi Dam	Madhya Pradesh	Narmada River
Bansagar Dam	Madhya Pradesh	Sone River
Gandhi Sagar Dam	Madhya Pradesh	Chambal River
Koyna Dam	Maharashtra	Koyna River
Indravati Dam	Odisha	Indravati River
Hirakud Dam	Odisha	Mahanadi River
Mettur Dam	Tamil Nadu	Kaveri River
Govind Ballabh Pant Sagar Dam also Rihand Dam	Uttar Pradesh	Rihand River
Tehri Dam	Uttarakhand	Bhagirathi River

RESERVOIRS

Himayat Sagar Reservoir	Telangana	Osman Sagar
Gobind Sagar Reservoir	Himachal Pradesh	Sutlej River
Maharana Pratap Sagar Reservoir	Himachal Pradesh	Pong Dam Lake
Salal Project	Jammu and Kashmir	Chenab River
Chutak Hydroelectric Project	Jammu and Kashmir	
Indirasagar Project	Madhya Pradesh	Narmada River
Narmada Dam Project	Madhya Pradesh	Narmada River
Rihand Project	Uttar Pradesh	Rihand River and Son River

THE CLIMATE OF INDIA

- India has tropical monsoon type of climate. It is greatly influenced by the presence of Himalayas in the North as they block the cold masses from Central Asia. It is because of Himalayas that the monsoons shed their water in India.
- **The Tropic of Cancer** (23.5°N) divide India into two almost equal climatic zones, namely; the Northern zone (sub-tropical) and the Southern zone (tropical).